HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

Explainable AI in Bioinformatics

ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Explainable AI in Bioinformatics

Practicing and Coding

ONE LOVE. ONE FUTURE.

Content

1. Overview
2. Applications
3. Case studies

Overview

- **Why interpretability matters in bioinformatics**
  - High-stakes decisions: diagnosis, drug discovery
  - Regulatory reasons (FDA, GDPR compliance)
  - Trust and adoption by biologists/clinicians
- **Types of interpretability**
  - **Intrinsic** (the model itself is interpretable, e.g., decision trees, linear models)
  - **Post-hoc** (explaining black-box models after training, e.g., SHAP, LIME)
- **Tradeoff between Accuracy and Interpretability**

- **Simple, interpretable models:**
  - Logistic Regression, Decision Trees
  - Rule-based models (e.g., RuleFit, RIPPER)
- **Post-hoc interpretation techniques:**
  - **Feature importance** (e.g., permutation importance)
  - **Local explanations** (e.g., LIME)
  - **Global explanations** (e.g., feature interaction maps)
  - **SHAP values** (SHapley Additive exPlanations)
- **Visualization tools**
  - Partial Dependence Plots (PDP)
  - Individual Conditional Expectation (ICE) plots
  - Feature attribution heatmaps

Applications

- **Gene expression analysis**:
  - Interpreting gene importance in classification (e.g., cancer type prediction)
- **Single-cell RNA-seq**:
  - Identifying key markers from ML models
- **Biomarker discovery**:
  - Feature selection + interpretation for disease biomarkers
- **Predicting protein structures or functions**:
  - Attention maps in protein contact prediction (e.g., AlphaFold interprets internal attention)
- **Drug response prediction**:
  - Feature interpretation in models predicting patient response to treatments

Case studies

- **Case Study 1**: Interpreting a Random Forest model predicting breast cancer subtype from gene expression data (using SHAP)
- **Case Study 2**: Using decision trees to identify patient subgroups based on genomic mutations
- **Case Study 3**: Interpreting deep learning models in pathology image analysis (CAMs, Grad-CAMs)

Challenging

- **Biological noise vs signal**: Interpretability needs caution.
- **Correlations ≠ Causations**: "Important" features might not be causal.
- **Data biases and model artifacts**.
- **Emerging solutions**:
  - Causal inference
  - Interpretable deep models (e.g., ProtoPNet, Concept Bottleneck Models)
- **Ethics and Responsible AI in Bioinformatics**

Coding

- https://drive.google.com/drive/folders/10wu4Zma7mPvLBxidykQdIVvN2si2bVXO?usp=sharing

HUST

Thank you

hust.edu.vn fb.com/dhbkhn